# CLI Eyeball Rescue

CS-20200 Term Project Proposal - Requirements Document

Project Title: CLI Eyeball Rescue

## Overview

This project is a command-line escape game played in the terminal. The user plays as a character named Kim Eyeball, who must rescue eyeballs captured by a Hunter and trapped inside a bottle. The goal of the game is to find the bottle of captured eyeballs in the dungeon and then escape through the exit while carrying the bottle.

At the beginning of the game, the user selects a difficulty. Easy mode uses a 5 x 5 map, Normal mode uses a 7 x 7 map, and Hard mode uses a 9 x 9 map. Each difficulty has one predefined fixed map. After the user selects a difficulty, the game loads the fixed map for that difficulty. Each map contains walls, an exit, a bottle of captured eyeballs, and hidden traps. Hidden traps are not directly shown on the screen. Instead, the game displays the number of hidden traps in each row and each column so that the user can infer possible trap locations.

## Storyline

Kim Eyeball had been living diligently at KAIST in harmony with the eyeballs. One day, a mysterious Hunter appeared, captured the eyeballs, trapped them inside a bottle, and locked them inside a dungeon. To rescue the captured eyeballs, Kim Eyeball sneaks into the dungeon. In a dungeon filled with hidden traps, can Kim Eyeball safely rescue the eyeballs?

## Symbols Used in the Game

| Symbol | Meaning |
|---|---|
| P | The player character, Kim Eyeball |
| B | The bottle containing the captured eyeballs |
| E | Exit |
| # | Wall |
| . | Empty tile or a hidden trap tile shown as empty |

## Requirements

### 1. Game Start and Difficulty Selection

- When the game starts, it prints the title and a short description of the game.
- At the beginning of the game, the user selects a difficulty.
- If the user enters 1, Easy difficulty is selected.
- If the user enters 2, Normal difficulty is selected.
- If the user enters 3, Hard difficulty is selected.
- If the user enters any value other than 1, 2, or 3 during difficulty selection, the game prints an error message and asks for input again.

### 2. Map Rule

- Easy difficulty has one predefined fixed 5 x 5 map.

- Normal difficulty has one predefined fixed 7 x 7 map.
- Hard difficulty has one predefined fixed 9 x 9 map.
- When the user selects a difficulty, the game loads the single predefined map for that difficulty.
- The selected map is used until the current game ends.
- If the user loses and chooses to restart, the game restarts with the same difficulty and reloads the fixed predefined map for that difficulty.
- Each map contains one player start position, one bottle of captured eyeballs, one exit, walls, and hidden traps.

## 3. Map Display and Trap Hints

- After a difficulty and map are selected, the game prints the selected map in the terminal.
- The map is displayed as a square grid.
- The player's current position is displayed as P.
- The bottle of captured eyeballs is displayed as B.
- The exit is displayed as E.
- Walls are displayed as #.
- Empty tiles are displayed as .
- Hidden traps exist in the internal map data, but they are displayed as empty tiles . on the screen.
- The game displays the number of hidden traps in each row to the right side of that row.
- The game displays the number of hidden traps in each column below the map.
- The displayed row and column trap counts must match the hidden trap locations in the selected predefined map.
- After every valid move, the game prints the updated map.

## 4. Player Movement

- The user moves by entering W, A, S, or D.
- W moves the player one tile up.
- A moves the player one tile left.
- S moves the player one tile down.
- D moves the player one tile right.
- Movement commands are case-insensitive, so w, a, s, and d are also accepted.
- Each valid movement command moves the player by exactly one tile.
- If the user enters an invalid movement command, the game prints an error message and asks for input again.
- Invalid input does not change the player's position.
- If the user tries to move outside the map, the player's position does not change and the game prints an error message.
- If the user tries to move into a wall tile #, the player's position does not change and the game prints an error message.

## 5. Bottle Rescue Rule

- The player starts the game without carrying the bottle.

- If the user reaches the bottle tile B, the game prints a message saying that the bottle of captured eyeballs has been found.
- After the bottle is found, the game records that the player is carrying the bottle.
- After the bottle is found, the bottle tile is treated as an empty tile.
- The player cannot win before finding the bottle.

### 6. Exit Rule

- If the user reaches the exit E before finding the bottle, the game prints a message saying that the player cannot escape without the captured eyeballs.
- If the user reaches the exit before finding the bottle, the game does not end.
- If the user reaches the exit E after finding the bottle, the user wins.
- When the user wins, the game prints a victory message and ends.

### 7. Hidden Trap and Restart Rule

- Hidden traps are not directly displayed on the map.
- Hidden traps appear as empty tiles . on the screen.
- If the user moves onto a hidden trap tile, the user loses immediately.
- If the user loses, the game prints a message saying that the player fell into the Hunter's hidden trap.
- After losing, the game asks whether the user wants to restart.
- If the user enters Y, the game starts a new game with the same difficulty and reloads the fixed predefined map for that difficulty.
- If the user enters N, the game ends.
- If the user enters any value other than Y or N during the restart choice, the game asks for input again.

### 8. Game Ending Conditions

- The game ends with a win when the user finds the bottle and then reaches the exit.
- The game ends when the user steps on a hidden trap and chooses N at the restart prompt.
- There is no tie condition in this game.

## Example Interaction

When the game starts, the user selects a difficulty. If the user enters 1, Easy difficulty is selected, and the game loads the predefined 5 x 5 Easy map. Hidden traps are not directly displayed and appear as empty tiles. The row trap count is displayed to the right side of each row, and the column trap count is displayed below the map.

```
CLI Eyeball Rescue
Select difficulty:
1. Easy (5 x 5)
2. Normal (7 x 7)
3. Hard (9 x 9)
Enter difficulty: 1

P . . # . row traps: 0
. # . . . row traps: 1
. . B . . row traps: 0
. # . . . row traps: 0
. . . . E row traps: 1
col traps:
1 0 0 1 0
```

```
Enter move (W/A/S/D): S
```

If the user enters S and the tile below the player is movable, Kim Eyeball moves one tile down and the updated map is printed.

```
You found the bottle of captured eyeballs!
You cannot escape without the captured eyeballs.
You escaped with the eyeballs. You win!
You fell into the Hunter's hidden trap. Game over.
Restart? (Y/N)
```

If the user reaches the B tile, the first message is printed. If the user reaches E before finding B, the second message is printed and the game continues. If the user reaches E after finding B, the third message is printed and the game ends. If the user steps on a hidden trap, the fourth message is printed, and the game asks whether the user wants to restart. If the user enters Y, the fixed predefined map for the same difficulty is reloaded and the game restarts. If the user enters N, the game ends.